
APPLIED MATHEMATICAL RESEARCH PROJECT

ESMI 2026

INTRODUCTIONS

1. Name
2. Major/intended major
3. Find one thing in common (a shared hobby or sport, a shared favorite super hero, maybe you're both middle children!)
4. Find a silly difference (a vegetable one of you likes and one of you hates, different favorite colors)
5. What is math research?

Introduce your peer:

1. Your peer's name
 2. Their intended major
 - Person (right): Your common ground
 - Person (left): Your silly difference
-

LOGISTICS

Instructor: Laurel Newman
(lan66@cornell.edu)

Lab Time:

- 22 June – 5 Aug: M/W/F 11 AM – 1 PM
- **Location:** 146 Upson

OH/Advising:

- **Time:** T/R 3– 4 PM
 - **Location:** 431 Rhodes
 - **Additional availability:** by appointment with 24 hour advance notice for M-F from 9 AM – 5 PM
 - **Slack:** Reach me from 9 AM – 8 PM, slow responses on Sat/Sun
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Timeline:

- M 15 June: program starts, first week daily workshops
 - W 17 June: students finalize their research groups and submit their group contracts
- F 19 June: Juneteenth off
- M 22 June: students' summer courses begin, we now meet M/W/F.
 - Student submit their project proposals
 - Mondays: research skills and best practices
 - Wednesdays: work days
 - Fridays: work days
- F 24 July: students submit their draft research report and presentation. Submit a ``working draft" of poster.
- F 31 July: ``working draft" of final research report and presentation due. Final version of abstract and poster: poster will be printed out in Mann library
- W 5 Aug: students submit their final research report and presentation to ESMI Box Folder by start of class
- R 6 Aug: Summer Research Symposium

ESMI 2026 Orientation Schedule	Date	Time	Location	Activity	Items Needed For The Day
	15-Jun	9:00 am - 11.30 am	Upson Hall 142	Orientation	-Laptop
		11:30 am - 1:00 pm	Okenshields Dinning	Lunch	-Note taking materials
		1:00 pm - 3:30 pm	Upson Hall 142	Applied Math Lecture	-Snacks if needed
	16-Jun	9:00 am - 11:00 am	Upson Hall 116	Learning Skills Workshop	-Laptop
		10:30 am - 12:00 pm	Upson Hall 116	Syllabus Review	-Note taking materials
		12:00 pm - 1:00 pm	On your own	Lunch	-Class Syllabus
		1:00 pm - 3:30 pm	Upson Hall 116	Applied Math Lecture	-Snacks if needed
	17-Jun	9:00 am - 11:00 am	Upson Hall 116	Applied Math Lecture	-Laptop
		12:00 pm - 1:00 pm	On your own	Lunch	-Note taking materials
		1:00 pm - 2:30 pm	Upson Hall 116	Managing Anxiety Workshop	-Snacks if needed
		2:30 pm - 4.30 pm	Upson Hall 116	Applied Math Lecture	
	18-Jun	8.30 am - 4.00 pm	Meet RPCC Parking Lot	High and Low Ropes Course	-Wear comfortable clothes and shoes for the physical movement -Water bottle -Sun screen -Snacks if needed -EpiPen if needed
	19-Jun	Juneteenth Celebration (federal Holiday, no campus activities)			
	20-Jun	Optional 12:00 pm - 6.00pm	South Side Community Center 305 S. Plain Street		Ithaca’s primary Juneteenth celebration is hosted by the Southside Community Center, featuring a massive block party with live music, dancing, soul food, and arts and crafts. The festival, running from noon to 6:00 PM on Saturday, June 20, 2026, takes place at 305 S. Plain Street

GOALS

- Gain experience doing mathematical research



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- Learn tools to engage with real-world mathematical models



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- Work collaboratively to explore an open-ended problem



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 - Use mathematical and computational tools to answer questions
-

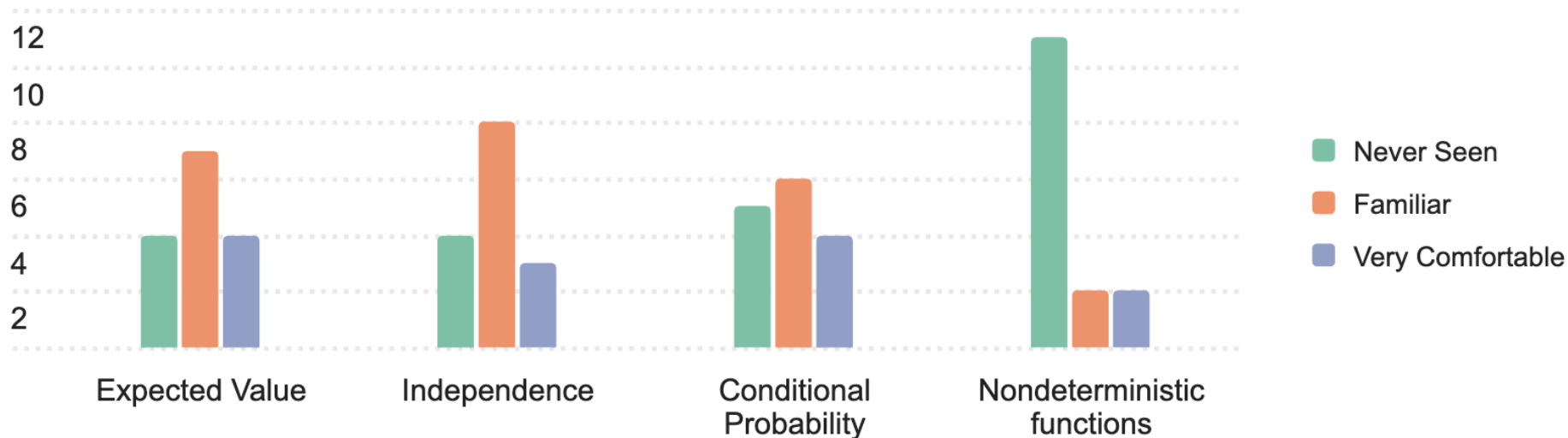
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 - Learn tools to engage with real-world mathematical models
 - Work collaboratively to explore an open-ended problem
 - Use mathematical and computational tools to answer questions
 - Formally write up and present results
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PRE-SURVEY RESPONSES

MATH CONCEPT EXPOSURE

Prior probability experience:



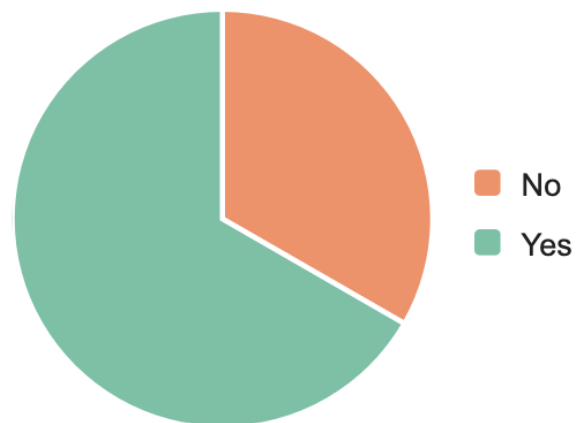
MATH CONCEPT EXPOSURE

Prior statistics experience:

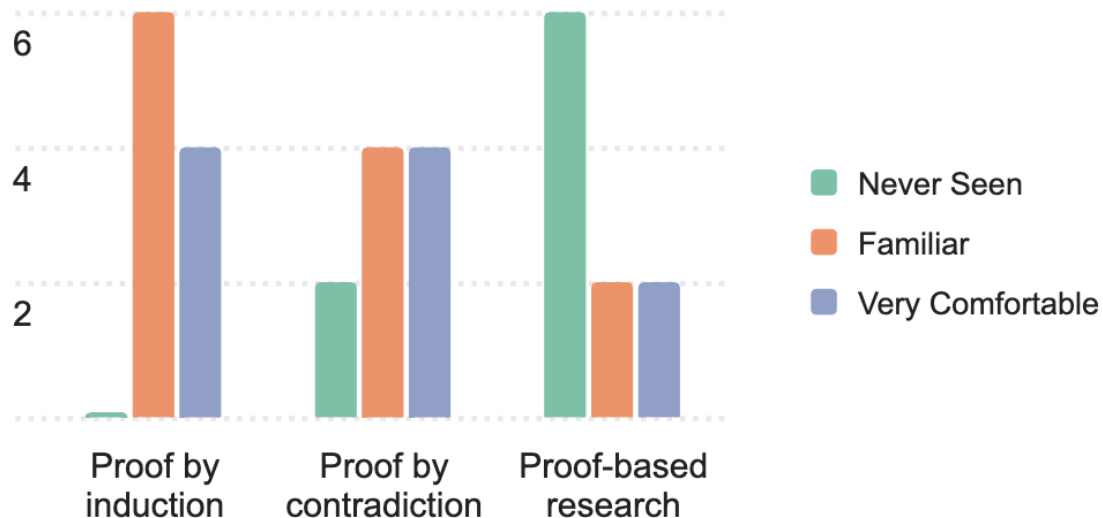


MATH CONCEPT EXPOSURE

Math Proofs?



Proof-Based concepts:



SCHEDULE FOR FIRST WEEK

Day	Topics	Length
Monday (today)	Research + Math Modelling	2.5 hr
Tuesday	Probability + Statistics	2.5 hrs
Wednesday	Python (simulation + graphing) + SQL + Tableau	2 hrs
Thursday	Ropes Course	-
Friday	Juneteenth	-

CODING BACKGROUND

Prior coding?

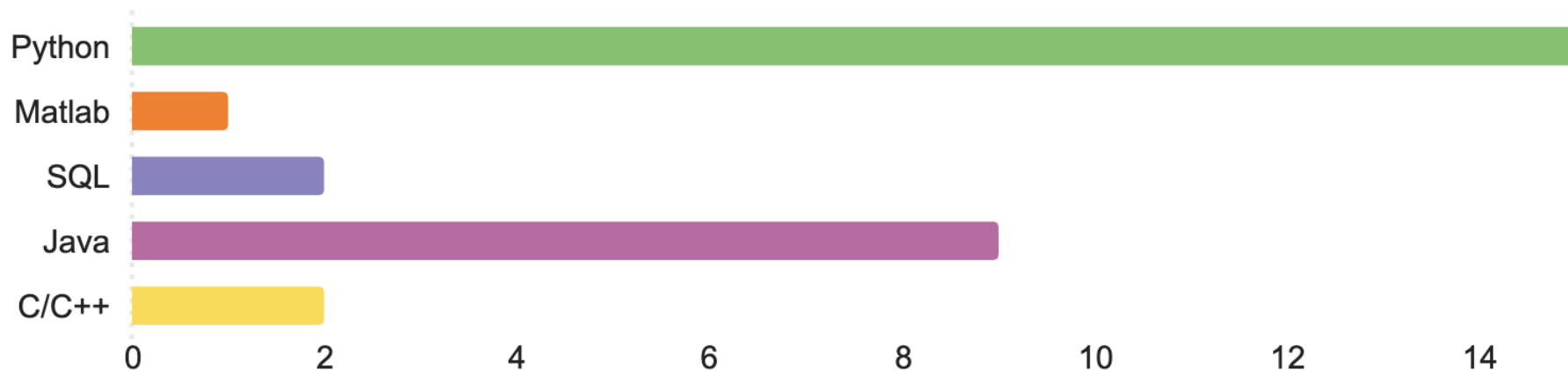


■ No
■ Yes

proficient
equivalent
python
inexperienced
2800 1112
cs2110
java
3120 development
complete
full ap
2110 code 1110
intermediate
cs1110 college
experienced

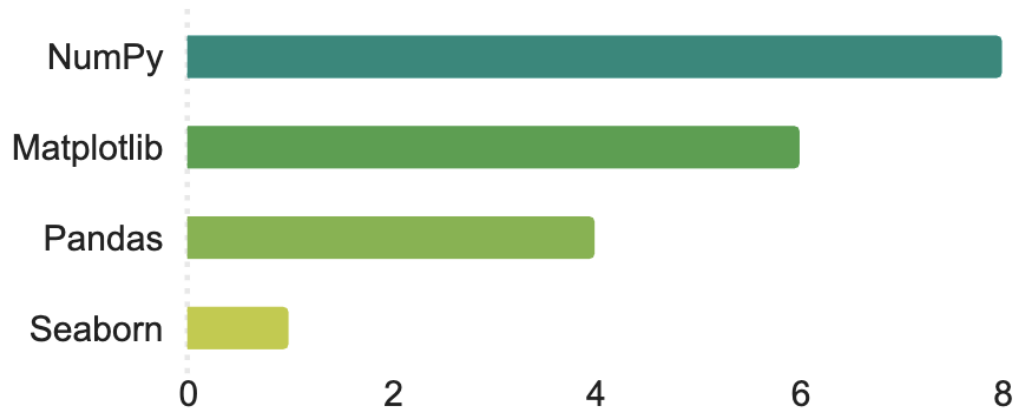
CODING BACKGROUND

Programming language familiarity:



CODING BACKGROUND

Python library familiarity:



SQL for data display
(e.g. Tableau)



NEXT 6 MONDAYS

- Finding Background + Literature Reviews
 - Coding Documentation + Presenting Results
 - Abstract Writing
 - Proofs
 - Research Posters + Talks + LaTeX
 - Writing up Results
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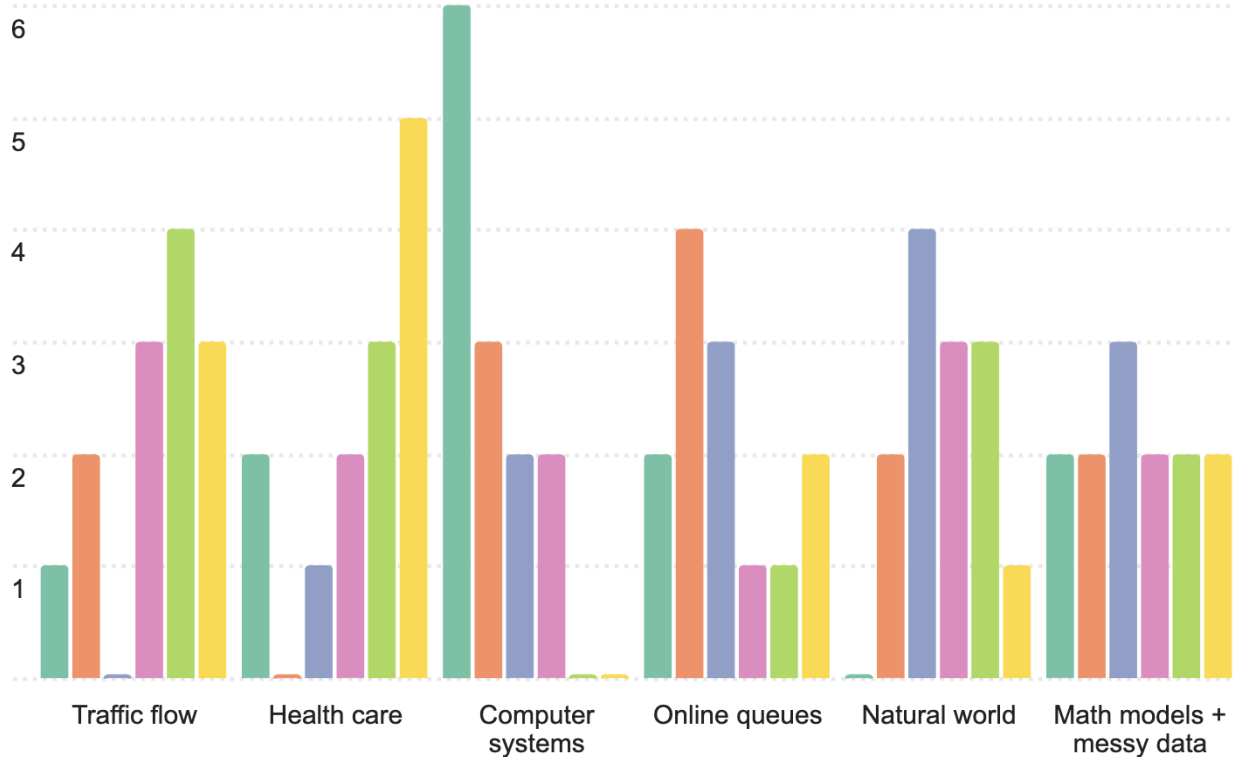
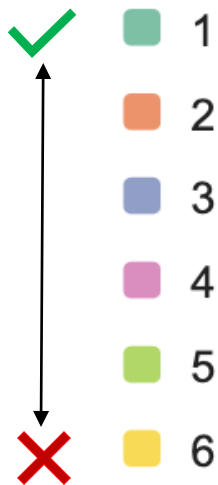
EXPECTATIONS

- Attend labs and actively engage with the material
 - Attendance is taken daily
 - Collaborate with a group of your classmates on a research problem
 - Come up with a real-world problem to model
 - Design a model
 - Test your model
 - Evaluate your model
 - Iterate!
-

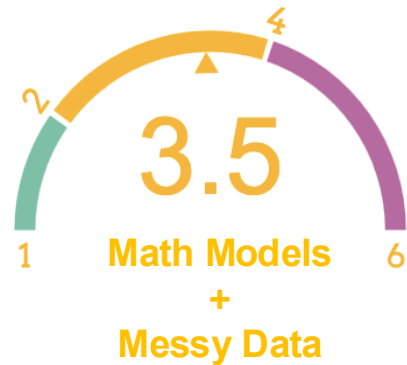
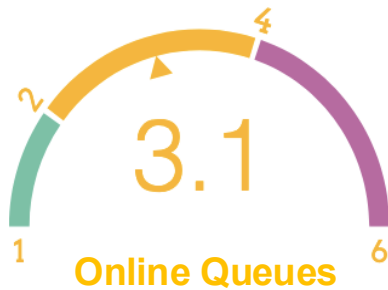
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 - Iterate!
 - Deliverables
 - Final research abstract + biographies (1 page)
 - Final paper (2-3 pages including abstract)
 - Final poster presentation
 - Elevator pitch (2 minutes, 2-3 slides)
 - Final oral presentation (15-20 minutes)
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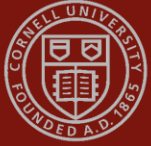
RESEARCH INTERESTS



RESEARCH INTERESTS



EXAMPLES: PRIOR ESMI RESEARCH



Modeling Microbial Crowding and Stability as a jammed state In M/M/1/K, FIFO, FILO Queue

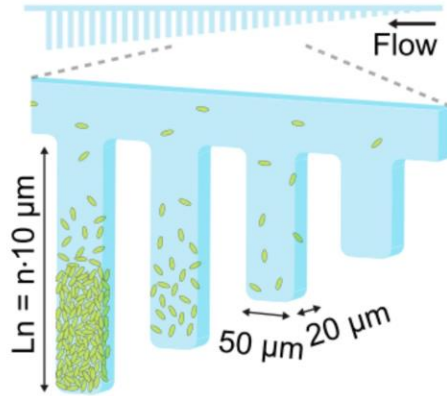


Figure 1: Microfluidic device from the paper with rectangles of varying size connected to a shared flow channel.

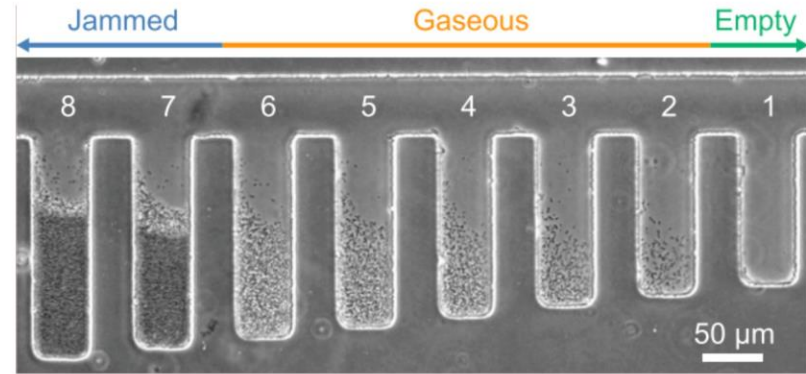
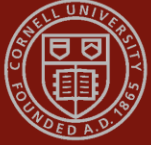
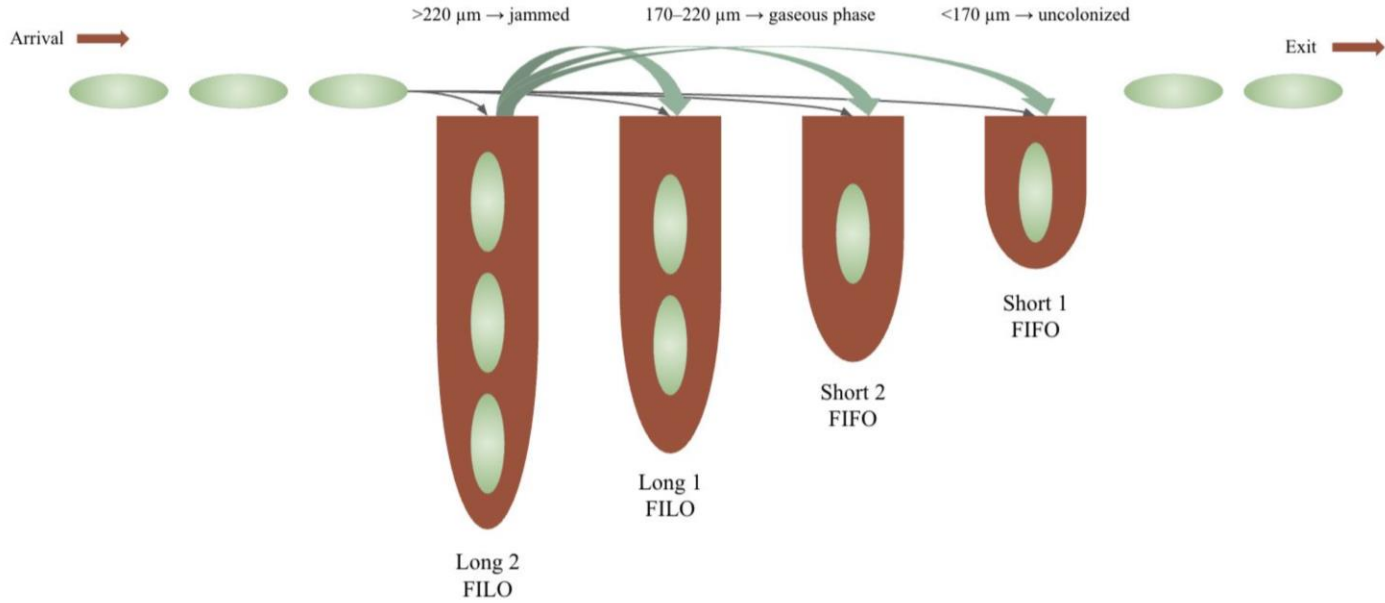


Figure 2: The steady state after 5 days of bacteria incubation

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Modeling Microbial Crowding and Stability as a jammed state In M/M/1/K, FIFO, FILO Queue



EXAMPLES: PRIOR ESMI RESEARCH



Markov Chain Modeling and Simulation of the North Atlantic Right Whale's Foraging Habits in Cape Cod Bay

Cornell
Engineering



Figure 1. North Atlantic Right whale schematic from NOAA Fisheries



Figure 2. Critical foraging habitat for the North Atlantic Right whale; zooplankton data was collected from Cape Cod Bay

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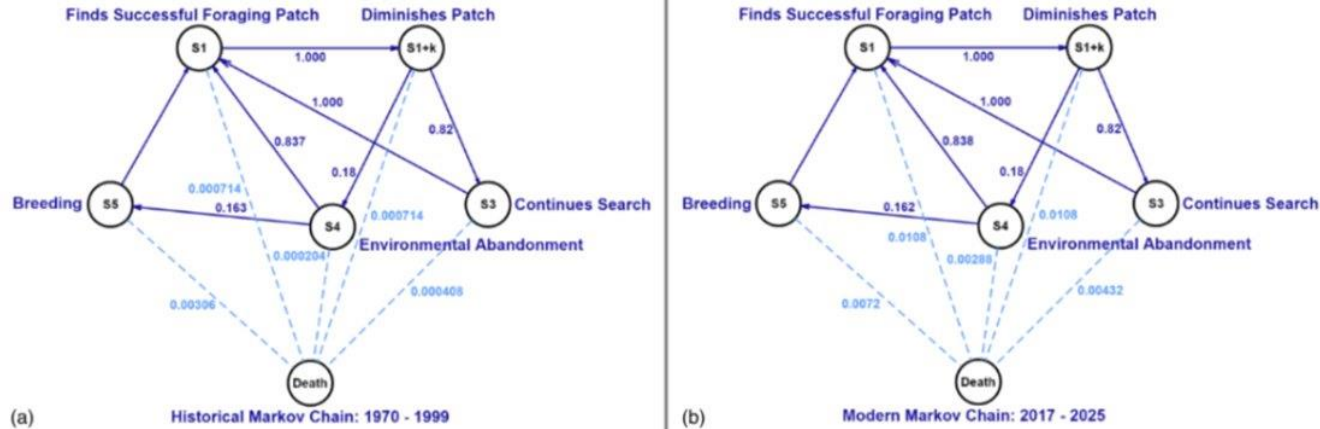
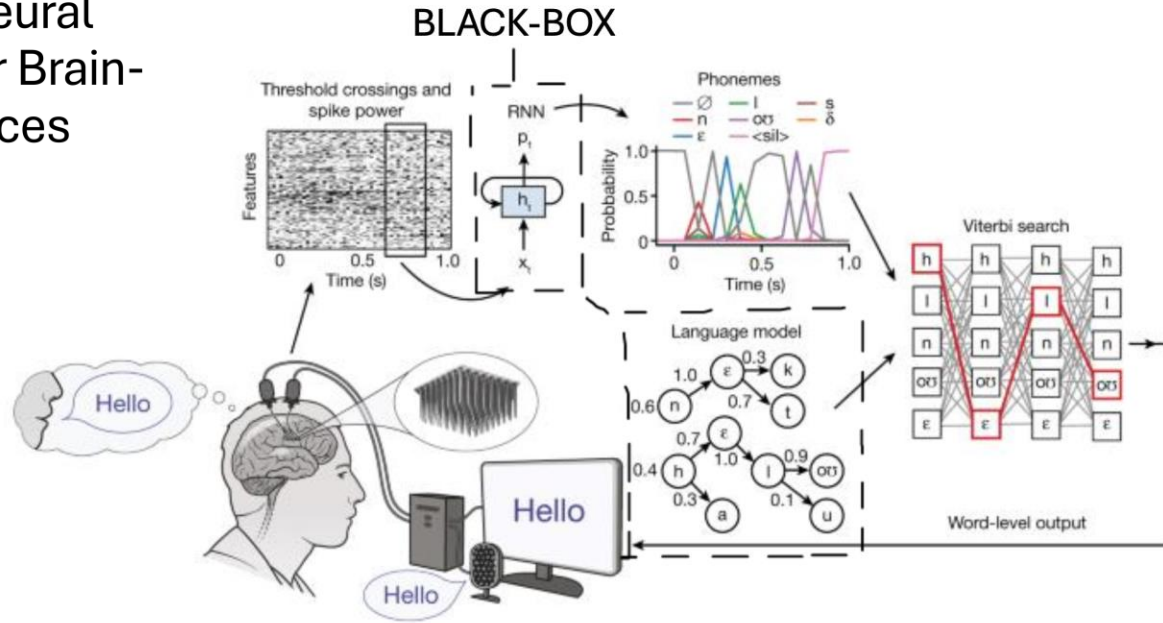


Figure 3. (a) Historical Markov Chain representing the behavioral states of a foraging North Atlantic Right Whale, with K = the max capacity of zooplankton aggregation (b) Modern model

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Integrating Priority-Based
Queueing Strategies into
Thought-to-Text Neural
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EXAMPLES

Integrating Priority-Based Queueing Strategies into Thought-to-Text Neural Decoding Pipeline for Brain-Computer Interfaces

